

Explainable Machine Learning for Fintech Adoption and Behavioural Segmentation in Pakistan

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Abstract—Using explainable machine learning (Random Forest and XGBoost with SHAP) and unsupervised segmentation (k-means), this study identifies who is excluded from Pakistan’s digital financial system and, more specifically, where formal access fails to translate into actual digital usage. Using the Global Findex Database 2025 (Pakistan, $n = 1,000$), we construct a four-level Adoption Tier target and test three hypotheses concerning the access-usage gap, the urban-rural adoption divide, and the relative importance of demographic versus access-based predictors. Random Forest achieves the best macro F1 (0.486) among four supervised models, and an independently-derived k-means cluster comprising 44.3% of the sample shows near-universal mobile phone ownership (98.0%) and majority internet use (62.3%), yet two-thirds remain formally excluded from the financial system – indicating that access is not the binding constraint for a substantial share of the population. Gender emerges as the strongest structural predictor of cluster membership (Cramer’s $V = 0.807$), exceeding the effect size of urban/rural status, diverging from prior Sub-Saharan African findings that access variables dominate adoption.

Index Terms—fintech adoption, financial inclusion, explainable AI, SHAP, k-means clustering, Pakistan, Global Findex

I. INTRODUCTION

Pakistan’s digital financial system has expanded rapidly over the past decade, yet the relationship between formal financial access and actual digital engagement remains poorly understood at the individual level. Existing research on financial inclusion in emerging economies has tended to treat access and usage as functionally equivalent, measuring inclusion primarily through account ownership rates. This conflation obscures a more granular and policy-relevant question: among people who do have access, who actually uses digital financial tools, and who does not?

This study applies explainable machine learning and unsupervised segmentation to individual-level survey data to identify who is excluded from Pakistan’s digital financial system and, more specifically, where formal access fails to translate into actual digital usage. Using Random Forest and XGBoost classifiers with SHAP explainability alongside k-means clustering, the analysis tests three hypotheses concerning the access-usage gap, the urban-rural adoption divide, and the relative importance of demographic versus access-based predictors of adoption.

II. LITERATURE REVIEW

Three recent studies inform the design of this project. Chowdhury et al. [1] applied Random Forest, XGBoost, and

k-means clustering to financial literacy and behaviour data from 1,067 adults in Bangladesh, validating their segmentation with silhouette, Davies-Bouldin, and Calinski-Harabasz indices. They identified three behavioural clusters (34%, 41%, and 25% of the sample) and reported a weak correlation between financial knowledge and financial behaviour, alongside a counterintuitive 7.3 percentage-point rural literacy advantage over urban respondents, with formal education emerging as a comparatively weak predictor relative to institutional trust.

Emuron, Kapim Kenfack, and Msimanga [2] took a more applied approach, predicting fintech adoption in South Africa from Findex, mobile pricing, and central bank data using regression-based methods on remittances, borrowing, age, and digital usage variables. Their work established a directly comparable variable set to the one constructed here from Findex Pakistan, though without a segmentation component.

Saidy and Hassan [3] modelled fintech adoption drivers across 36 Sub-Saharan African countries using the 2021 Global Findex round, combining Logit and Probit regression with Random Forest, Gradient Boosting, and XGBoost. Their central finding was that formal account ownership and technology access – specifically mobile phone ownership and internet connectivity – were the strongest predictors of adoption.

This project combines Chowdhury et al.’s dual supervised-unsupervised methodology with the adoption-prediction framing of Emuron et al. and Saidy and Hassan, applying both to Pakistan using the newer 2024-fieldwork Findex round [4] – a combination not yet applied to South Asian data, and a direct test of whether the Sub-Saharan African finding that access dominates adoption generalizes to a different regional context.

III. DATA AND METHODS

A. Data

The analysis uses the Global Findex Database 2025, Pakistan microdata (Reference: PAK_2024_FINDEX_v02_M) [4], an individual-level survey of 1,000 respondents conducted by the World Bank between May and December 2024. The dataset includes 183 variables spanning demographics, account ownership, digital and informal financial behaviour, and financial resilience.

B. Target Construction

A four-level ordinal Adoption Tier was constructed from account ownership and digital usage indicators: Excluded (no formal account), Basic (account held, no digital usage), Digitally Engaged (uses digital payments and/or a digitally enabled account), and Advanced (digital usage combined with formal saving or borrowing behaviour). The resulting distribution was 68.2% Excluded, 2.1% Basic, 10.1% Digitally Engaged, and 19.6% Advanced, shown in Fig. 1.

C. Feature Selection and Leakage Control

Twenty variables across four construct groups (demographics, access, behaviour, resilience) were selected from the Findex codebook. Because the Adoption Tier target was constructed directly from `account`, `dig_account`, `anydigpayment`, `saved`, and `fin22a` – and because `account_fin` and `account_mob` are near-deterministic components of `account` itself – these seven variables were excluded from the supervised and unsupervised feature sets. An initial model run that retained these variables produced spurious near-perfect accuracy, which was identified as data leakage and corrected before further analysis; this correction is reported for methodological transparency.

D. Supervised Modelling

Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers were trained on a stratified 75/25 train-test split to predict Adoption Tier from the remaining fifteen features, with class weighting applied to address the imbalanced target distribution. Models were evaluated on accuracy and macro/weighted F1, and the best-performing model was explained using SHAP.

E. Unsupervised Segmentation

K-means clustering was applied to the same leakage-safe feature set (standardized via z-score) for $k = 2$ to 5, with k selected by silhouette score and Davies-Bouldin and Calinski-Harabasz indices computed for robustness (Fig. 5). Clusters were profiled on demographic and access characteristics and visualized via a two-component PCA projection (Fig. 6).

F. Statistical Validation

Chi-square tests of independence with Cramer’s V effect sizes were used to test associations between cluster membership, Adoption Tier, gender, and urban/rural status. Wilson score 95% confidence intervals were computed for all headline proportions, given the small size of some subgroups (e.g. $n = 21$ for the Basic tier).

IV. RESULTS

A. H1 – The Access-Usage Gap

Among the 318 respondents holding a formal account, 6.6% (95% CI: 4.4%–9.9%) showed no digital financial usage – a direct measure of the dormant-account phenomenon described by Chowdhury et al. [1] as a knowledge-behaviour gap in the Bangladesh context.

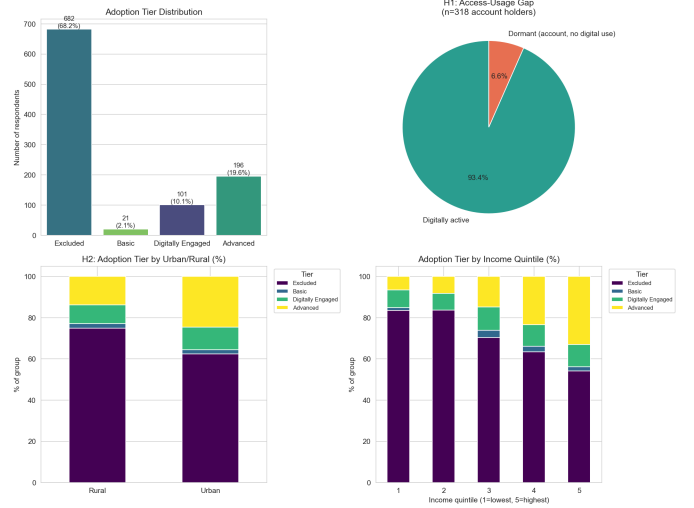


Fig. 1. Adoption Tier distribution, H1 access-usage gap, and Adoption Tier by urban/rural and income quintile.

A substantially larger and more striking pattern emerged from the independent, unsupervised cluster analysis. K-means ($k = 3$, selected by silhouette score, though the margin over $k = 2$ was narrow: 0.1675 vs. 0.1674, see Fig. 5) identified a cluster – labelled “Connected but Unengaged” – comprising 44.3% of the entire sample ($n = 443$). This cluster shows near-universal mobile phone ownership (98.0%) and majority internet use (62.3%), yet 66.8% (95% CI: 62.3%–71.0%) remain formally Excluded from the financial system. Because this pattern emerged from clustering on access and demographic variables without reference to the Adoption Tier labels, it constitutes independent confirmation that access is not the binding constraint for a large share of the population – a considerably stronger and more generalizable version of H1 than the direct dormant-account statistic alone (Table II).

B. H2 – Urban versus Rural Adoption

Adoption Tier was significantly associated with urban/rural status ($\chi^2 = 21.742$, $p = 0.0001$, $df = 3$, Cramer’s V = 0.147), and cluster membership showed a parallel association ($\chi^2 = 19.823$, $p < 0.0001$, $df = 2$, Cramer’s V = 0.141). The direction of the effect favoured urban respondents: rural share declined monotonically across clusters of increasing adoption, from 51.0% in the least-adopted cluster to 46.0% and then 28.6% in progressively more digitally engaged clusters. This is consistent with the conventional urban-adoption advantage reported broadly in the financial inclusion literature, and contrasts with Chowdhury et al.’s reversed finding of a rural literacy advantage in Bangladesh [1] – suggesting the rural advantage identified there does not generalize to behavioural adoption in the Pakistani context.

C. H3 – Predictors of Adoption

Model comparison results (Table I) show XGBoost achieved the highest accuracy (0.800) but Random Forest achieved the highest macro F1 (0.486), the metric prioritized here given

TABLE I
SUPERVISED MODEL COMPARISON

Model	Accuracy	F1 (macro)	F1 (weighted)
XGBoost	0.800	0.481	0.781
Random Forest (best)	0.708	0.486	0.737
Decision Tree	0.488	0.424	0.554
Logistic Regression	0.420	0.392	0.473

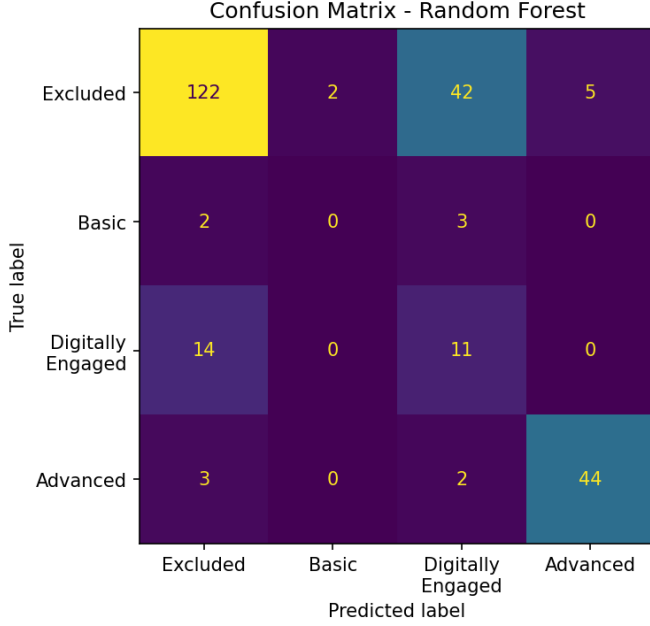


Fig. 2. Confusion matrix for the best-performing model (Random Forest).

substantial class imbalance in the target. All models struggled with the Basic tier ($n = 21$), which is attributable to sample size rather than model deficiency (Fig. 2).

SHAP analysis of the Random Forest model (Fig. 4) ranked predictors as follows: saving via mobile money, workforce participation, informal saving, gender, mobile phone ownership, age, internet use, and income quintile. Notably, mobile phone ownership and internet use – the two strongest predictors identified by Saidy and Hassan across Sub-Saharan Africa [3] – ranked in the middle of the importance distribution rather than at the top, suggesting that in Pakistan, existing financial behaviour (formal and informal saving patterns) carries more predictive weight than raw technology access once other variables are controlled for. Gender’s persistence in the top four SHAP-ranked features, after controlling for age, income, and workforce status, indicates its association with adoption is not merely confounded by these other demographic factors.

This gender effect was confirmed with substantially larger magnitude in the cluster analysis. Cluster membership was very strongly associated with gender ($\chi^2 = 650.549$, $p < 0.0001$, $df = 2$, Cramer’s $V = 0.807$), and Adoption Tier itself showed a large association with gender ($\chi^2 = 162.080$, $p < 0.0001$, $df = 3$, Cramer’s $V = 0.403$) – both well above the effect sizes observed for urban/rural status. The

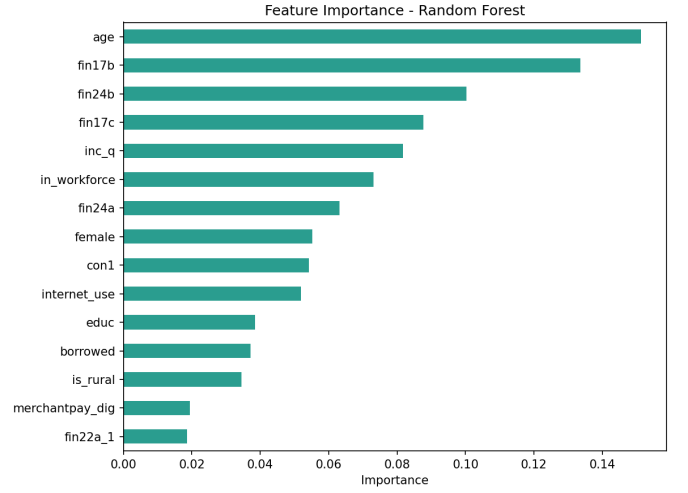


Fig. 3. Random Forest feature importance (Gini-based).

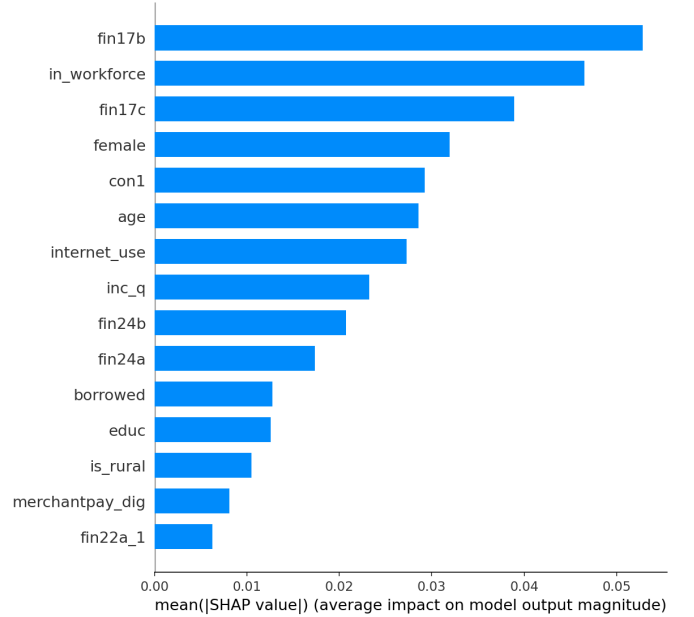


Fig. 4. Mean absolute SHAP value per feature, averaged across all four Adoption Tier classes.

most digitally excluded cluster was 96.3% female (95% CI: 94.1%–97.7%), compared to 16.3% and 10.3% female in the two more digitally engaged clusters (Table II). Combined with the PCA loadings (Fig. 6), which show gender and access variables loading on the same principal component, this indicates that in this sample, gender and digital access are structurally intertwined rather than independent predictors of adoption.

V. DISCUSSION

The central contribution of this analysis is the identification of a large, access-rich but adoption-poor population segment, discovered independently through unsupervised clustering. At 44.3% of the sample, this segment is far larger than the

TABLE II
CLUSTER PROFILES (K-MEANS, K=3)

Cluster	<i>n</i> (%)	% Female	% Rural	% Mobile	% Excluded
0 – Digitally Excluded Women	431 (43.1%)	96.3%	51.0%	28.3%	89.3%
1 – Connected but Unengaged	443 (44.3%)	16.3%	46.0%	98.0%	66.8%
2 – Advanced Adopters	126 (12.6%)	10.3%	28.6%	98.4%	0.8%

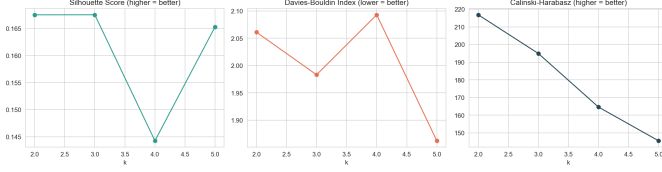


Fig. 5. Silhouette, Davies-Bouldin, and Calinski-Harabasz scores for $k = 2$ to 5.

6.6% dormant-account rate would suggest, and it reframes the practical question facing Pakistan’s digital finance sector: the primary obstacle to inclusion is not, for a substantial share of the population, a lack of phones or internet connectivity, but something downstream of that access – plausibly trust, product design, documentation requirements, or social/institutional barriers not captured directly in this dataset.

The gender findings compound this picture. Rather than a marginal demographic control variable, gender shows one of the largest effect sizes in the entire analysis (Cramer’s $V = 0.807$ for cluster membership), exceeding even the urban/rural divide by a wide margin. This diverges from Saidy and Hassan’s Sub-Saharan African findings, where account ownership and technology access dominated [3], and suggests that regional context materially changes which factors predict fintech adoption – access-centric interventions calibrated for Sub-Saharan Africa may be poorly suited to Pakistan, where the access-usage gap and the gender gap appear to be entangled with each other rather than separate problems requiring separate interventions.

The H2 finding – urban adoption exceeding rural adoption – aligns with the mainstream financial inclusion literature but diverges from Chowdhury et al.’s Bangladesh result [1]. Since Chowdhury et al. measured financial literacy rather than behavioural adoption, this divergence may reflect a genuine difference between knowledge and behaviour as outcome variables, rather than a contradiction between the two country contexts.

VI. LIMITATIONS

- The Basic tier ($n = 21$) is too small for reliable classification by any model tested; this is a sample-size constraint rather than a modelling failure and should not be over-interpreted.
- The variable `fin17b` (saved via mobile money), the top-ranked SHAP predictor, is partly confounded with mobile money account ownership due to Findex’s survey routing logic (approximately 74% of respondents were routed out of this question); its top ranking should be interpreted with this caveat.

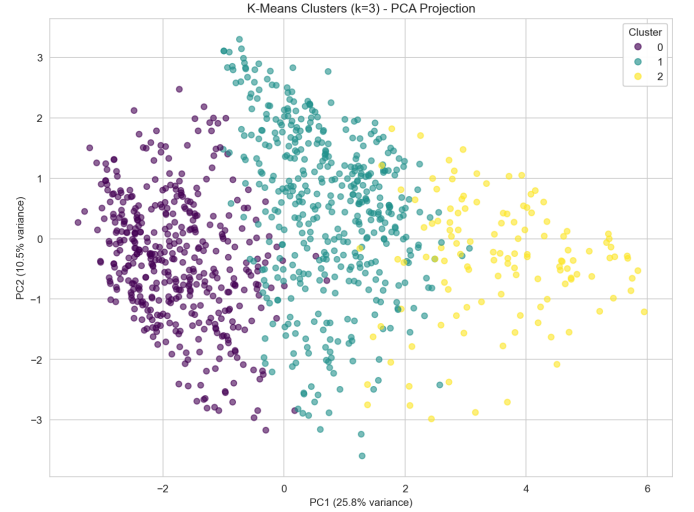


Fig. 6. PCA projection of k-means clusters ($k = 3$).

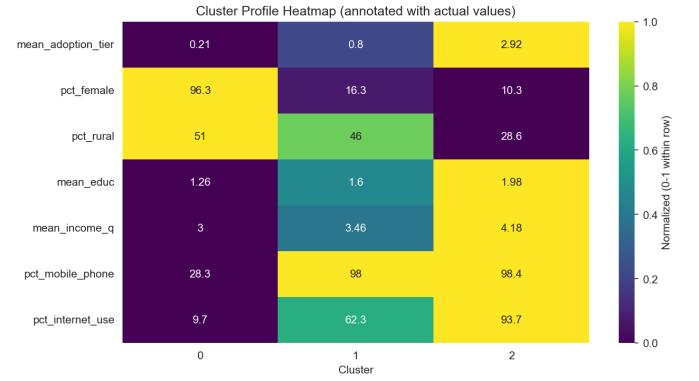


Fig. 7. Normalized cluster profile heatmap across demographic and access variables.

- The silhouette score for $k = 3$ was only marginally higher than for $k = 2$, indicating the three-cluster solution, while interpretable, is not an overwhelmingly dominant statistical structure.
- Several chi-square tests involve subgroups with small expected cell counts (notably comparisons involving the Basic tier); results for these specific comparisons should be treated as indicative rather than definitive.
- All findings are associational. Findex is a cross-sectional survey, and no causal claims are made regarding the drivers of adoption or the mechanisms underlying the gender and access-usage gaps identified.

VII. FUTURE WORK

Three extensions would strengthen this analysis. First, replicating the pipeline on the Financial Inclusion Insights (FII) Bangladesh tracker survey, if obtained, would provide a genuine same-instrument cross-country comparison. Second, a qualitative or mixed-methods follow-up focused specifically on the “Connected but Unengaged” cluster could clarify which downstream barriers – trust, documentation, product design

– best explain non-adoption despite access. Third, bootstrap resampling of the train-test split would provide a confidence interval on model performance metrics, strengthening the robustness claims made about the SHAP-based driver ranking.

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